

Shih-Hung Wei 魏士閔

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Summary

Master of Engineering in Computer Science graduate from Virginia Tech with hands-on experience in building intelligent AI systems, scalable backend services, and full-stack applications. Skilled in Python, Java, and C++ with deployments on AWS, GCP, and Azure. Passionate about leveraging multi-agent systems and LLM technologies to build impactful, production-ready software solutions. Currently working at TSMC and seeking opportunities to drive innovation in software and AI development.

Education

Virginia Polytechnic Institute and State University(Virginia Tech), Falls Church, VA 2023 – 2024
Master of Engineering - Computer Science

National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan 2018 – 2022
Bachelor of Science, Department of Computer Science
Bachelor of Science, Department of Biological Science and Technology

Skills

General Development	Python3, C/C++, Java, JavaScript, TypeScript, Git, Bash, AWS, GCP, Azure
Frameworks	React, Next.js, Node.js, Django, Firebase, PyTorch, Scikit-learn, TensorFlow
Data & Infrastructure	Spark, Hadoop, Docker, Kubernetes, RESTful APIs, RAG
Interested Fields	AI Agents, Multi-Agent Systems, Frontend, Backend, Machine Learning

Work Experience

Software Engineer, TSMC, Hsinchu, Taiwan Mar.2025 – present.

- Developed and deployed Web RESTful APIs using Java for internal enterprise-level applications on Azure Kubernetes Service.
- Designed a text-to-SQL service with Retrieval-Augmented Generation (RAG) architecture for semiconductor manufacturing data, enabling non-technical users to query production databases through natural language.

Software Developer Intern, Radical AI, Remote, United States June.2024 – Aug.2024

- Contributed to **Kai**, an open-source AI Coach, by implementing enhancements and debugging using **Node.js** and **React.js**.

Research & Development Intern, SHOPLINE Technology Corp., Taipei, Taiwan July.2022 – Feb.2023

- Implemented operations quality requirements on **AWS**, to ensure SHOPLINE's e-commerce platform integrity and compliance.
- Identified 150+ defects and developed **JavaScript** test scripts, boosting product quality by 30% and front-end reliability.

Undergraduate Researcher, Drug Design and Systems Biology Laboratory(BioXGEM), NYCU July.2020 – June.2022

- Developed a full-stack web tool using JavaScript/PHP and Python for lab server utilities and moiety extractions, integrating multiple Python tools to improve data analysis workflow with over 30 daily uses.. | [link](#)

Selected Projects

CoverCraft AI - Multi-Agent Cover Letter System | [link](#) Dec.2024 – present.

- Built an AI system using Google **Agent Development Kit** (ADK) to automate cover letter generation.
- Implemented multi-agent architecture with context-driven state management across specialized agents with PDF processing, web research, web scraping, and data validation to generate ATS-optimized output with comprehensive error handling.

Airline Data Query Platform – Capstone Project (Master’s Degree) Aug.2024 – Dec.2024

- Built a web-based platform for analyzing U.S. domestic airline data using **Spark/Hadoop** for distributed processing with a singleton pattern, **Django** and **React.js** with **RESTful APIs** for interface and deployed via **Docker** on a **Kubernetes** cluster.
- Developed an intuitive natural language query interface leveraging an Large Language Model(LLM) to convert English requests into SQL queries, enhancing accessibility for non-technical users.

wei4r.type - Zhuyin Typing Game | [type.wei4r.com](#) Mar.2023 – June.2024

- Designed and deployed a Zhuyin typing game using Next.ts and Firebase with real-time performance metrics, hosted on Vercel with custom domain management.

BookShelf - Online Bookstore – Course Project Aug.2023 – May.2024

- Built a full-stack bookstore platform using React, JDBC, and AWS with secure VPC, EC2, and Docker configuration.